

assignment projectile

⚠ This is a preview of the published version of the quiz

Started: Apr 29 at 7:54pm

Quiz Instructions

2 attempts. Tutorial in the dropbox.



Question 1 10 pts

The x component of vector **A** is 5.3 units, and its y component is -2.3 units. The angle that vector **A** makes with the +x-axis is closest to

hint

from origin. run 5.3 @ right. Then fall 2.3 @ down. Connect the starting point to ending point. It makes a triangle. What is the angle with the +xaxis. See my tutorial. step by step

Use tangent.

☐

23°.

☐

340°.

☐

250°.

☐

160°.

☐

110°.



Question 2 10 pts

A car travels 20 km west and then 20 km south. What is the magnitude of its displacement vector?

from origin. walk 20 @ left. then 20 @ down. connect the starting point to end point. You get a triangle.

Use pythagorian theorem

☐

0 km

☐

40 km



20 km



28 km



Question 3 10 pts

A girl throws a rock horizontally, with a velocity of 10 m/s, from a bridge. It falls 20 m to the water below. How far does the rock travel horizontally before striking the water, assuming negligible air resistance?



20 m



16 m



24 m



14 m



Question 4 10 pts

You throw a rock horizontally off a cliff with a speed of 20 m/s and no significant air resistance. After 2s , the magnitude of the velocity of the rock is closest to

hint: V_x stays the same. $V_y=0$ at $t=0$ but $V_y=20\text{m/s}$ at $t=2\text{s}$. So find the magnitude with Pythagorean theorem



40 m/s.



37 m/s.



28 m/s.



20 m/s.



Question 5 10 pts

In an air-free chamber, a pebble is thrown horizontally, and at the same instant a second pebble is dropped from the same height. Compare the times of fall of the two pebbles.



The thrown pebble hits first.



The dropped pebble hits first.



We cannot tell without knowing which pebble is heavier.



They hit at the same time.



Question 6 10 pts

A stone is thrown horizontally with an initial speed of 10 m/s from the edge of a cliff. A stopwatch measures the stone's trajectory time from the top of the cliff to the bottom to be 4.3 s. What is the height of the cliff if air resistance is negligibly small?



43 m



77 m



91 m



22 m



Question 7 10 pts

A ball thrown horizontally from a point 24 m above the ground, strikes the ground after traveling horizontally a distance of 18 m. With what speed was it thrown, assuming negligible air resistance?



8.9 m/s



7.4 m/s



8.1 m/s



6.1 m/s



Question 8 10 pts

For general projectile motion with no air resistance, the horizontal component of a projectile's velocity



first decreases and then increases.



remains a non-zero constant.



continuously decreases.



continuously increases.



remains zero.



Question 9 10 pts

A ball is thrown with an initial velocity of 20 m/s at an angle of 60° above the horizontal. If we can neglect air resistance, what is the horizontal component of its instantaneous velocity at the exact top of its trajectory?

See my tutorial.

☐

zero

☐

20 m/s

☐

17 m/s

☐

10 m/s



Question 10 10 pts

For general projectile motion with no air resistance, the horizontal component of a projectile's acceleration

☐

is always zero.

☐

continuously increases.

☐

remains a non-zero constant.

☐

first decreases and then increases.

☐

continuously decreases.

Not saved

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